



**PHENIKAA UNIVERSITY**  
**Faculty of Computer Science**

# Software Analysis and Design

## Module 4: Business Process Modeling with Activity Diagram

- ❖ Business process modeling
- ❖ Elements of an activity diagram
- ❖ Creating activity diagram

# Business process modeling



- ❖ Business process models describe the different activities that, when combined, support a business process. Business processes typically cut across functional departments
- ❖ Business processes modeling is a very constructive activity that can be used to make sense of the gathered requirements
- ❖ Business process models are very powerful tools for communicating the analyst's current understanding of the requirements to the user.

# Best practices for business processes modeling



- ❖ Be realistic, because it is virtually impossible to identify everything that is included in a business process at this point in the evolution of the system.
- ❖ Be agile because even though we might not identify every single feature of a business process, the features that we do identify should be identified in a rigorous manner.
- ❖ All modeling is a collaborative/social activity. Therefore, business process modeling must be performed with teams, not by individuals.

# Best practices for business processes modeling



- ❖ Do not use a CASE tool to do the modeling but use whiteboards instead. However, once the process is understood, it is a good idea to use a CASE tool to document the process.
- ❖ Process modeling should be done in an iterative manner. As you better understand a business process, you will need to return to the documented version of the process and revise it.

# Best practices for business processes modeling



- ❖ When modeling a business process, stay focused on that specific process. If tasks associated with other business processes are identified, simply record them on a to-do list and get back to the business process that you are currently modeling.
- ❖ Remember that a business process model is an abstraction of reality. By that, we mean that you should not include every minor task in the current description of the business process.

- ❖ Business process modeling
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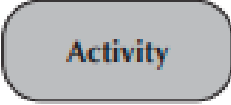
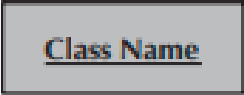
# Activity diagram



- ❖ Activity diagrams are used to model the behavior in a business process independent of objects. Activity diagrams can be used to model everything:
  - ❖ a high-level business workflow that involves many different use cases or
  - ❖ the details of an individual use case (thường là minh họa cho luồng sự kiện của UC)

# Elements of an activity diagram



|                                                                                                                                                                                             |                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>An action:</b> <ul style="list-style-type: none"><li>■ Is a simple, nondecomposable piece of behavior.</li><li>■ Is labeled by its name.</li></ul>                                       |    |
| <b>An activity:</b> <ul style="list-style-type: none"><li>■ Is used to represent a set of actions.</li><li>■ Is labeled by its name.</li></ul>                                              |    |
| <b>An object node:</b> <ul style="list-style-type: none"><li>■ Is used to represent an object that is connected to a set of object flows.</li><li>■ Is labeled by its class name.</li></ul> |    |
| <b>A control flow:</b> <ul style="list-style-type: none"><li>■ Shows the sequence of execution.</li></ul>                                                                                   |    |
| <b>An object flow:</b> <ul style="list-style-type: none"><li>■ Shows the flow of an object from one activity (or action) to another activity (or action).</li></ul>                         |    |
| <b>An initial node:</b> <ul style="list-style-type: none"><li>■ Portrays the beginning of a set of actions or activities.</li></ul>                                                         |   |
| <b>A final-activity node:</b> <ul style="list-style-type: none"><li>■ Is used to stop all control flows and object flows in an activity (or action).</li></ul>                              |  |
| <b>A final-flow node:</b> <ul style="list-style-type: none"><li>■ Is used to stop a specific control flow or object flow.</li></ul>                                                         |  |

# Elements of an activity diagram



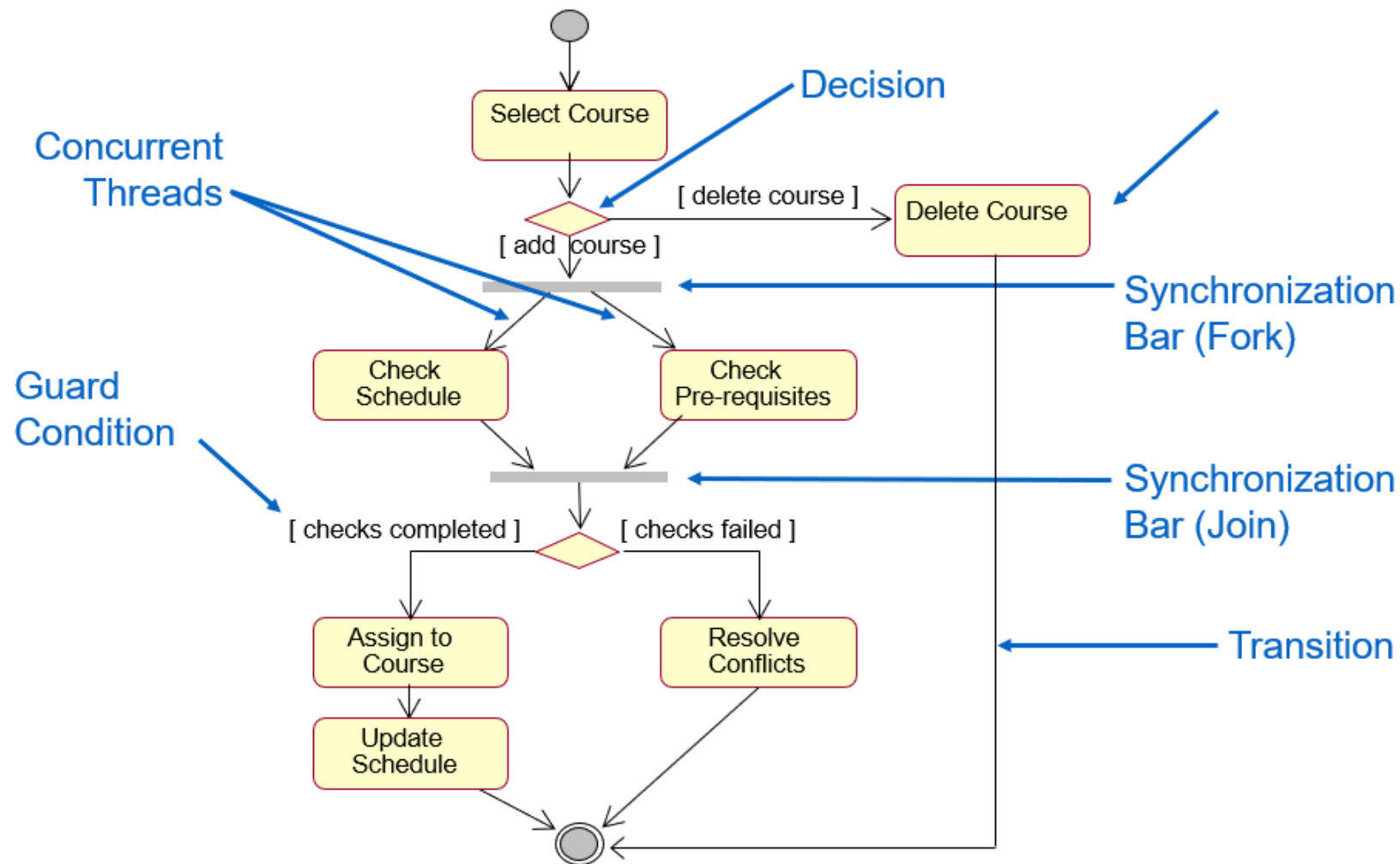
|                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>A decision node:</b></p> <ul style="list-style-type: none"><li>■ Is used to represent a test condition to ensure that the control flow or object flow only goes down one path.</li><li>■ Is labeled with the decision criteria to continue down the specific path.</li></ul>                                | A diamond-shaped node with an incoming arrow from the top. Two outgoing arrows branch out to the left and right, each labeled with "[Decision Criteria]" in blue text. |
| <p><b>A merge node:</b></p> <ul style="list-style-type: none"><li>■ Is used to bring back together different decision paths that were created using a decision node.</li></ul>                                                                                                                                    | A diamond-shaped node with two incoming arrows from the top-left and top-right. A single outgoing arrow points downwards from the bottom of the diamond.               |
| <p><b>A fork node:</b></p> <p>Is used to split behavior into a set of parallel or concurrent flows of activities (or actions)</p>                                                                                                                                                                                 | A horizontal bar with a single incoming arrow from the top. Two outgoing arrows branch out downwards from the bar.                                                     |
| <p><b>A join node:</b></p> <p>Is used to bring back together a set of parallel or concurrent flows of activities (or actions)</p>                                                                                                                                                                                 | A horizontal bar with two incoming arrows from the top. A single outgoing arrow points downwards from the center of the bar.                                           |
| <p><b>A swimlane:</b></p> <p>Is used to break up an activity diagram into rows and columns to assign the individual activities (or actions) to the individuals or objects that are responsible for executing the activity (or action)</p> <p>Is labeled with the name of the individual or object responsible</p> | A gray rectangular box with the word "Swimlane" written in black text inside.                                                                                          |

# Content



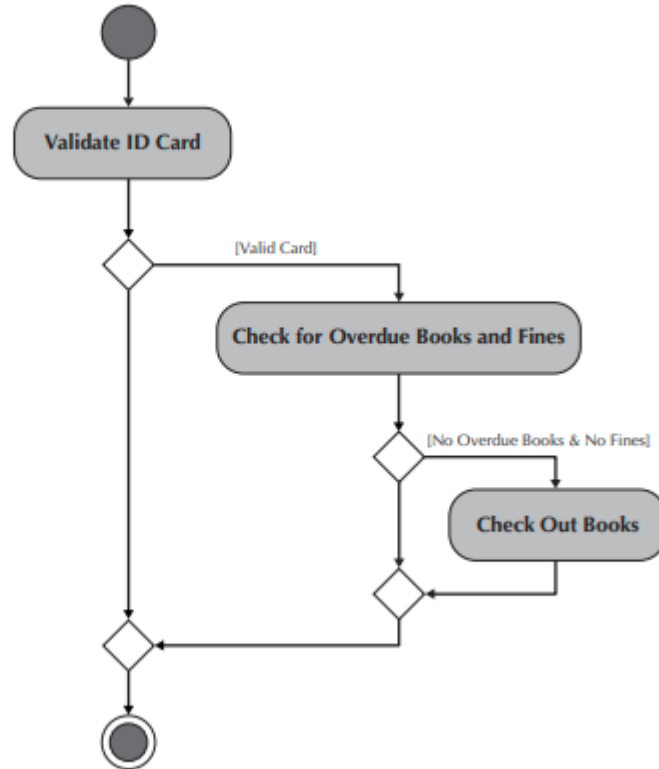
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# Activity diagram for UC: Register for courses



Lưu ý: Trong sơ đồ này không dùng merge node (để đơn giản có thể vẽ theo cách này)

# Activity diagram for UC: Borrow Books



**FIGURE 4-12**  
Activity Diagram for  
the Borrow Books  
Use Case

# Activity diagram

Activity diagram cho UC quản lý lịch  
hẹn (Trong hệ thống quản lý bệnh viện)

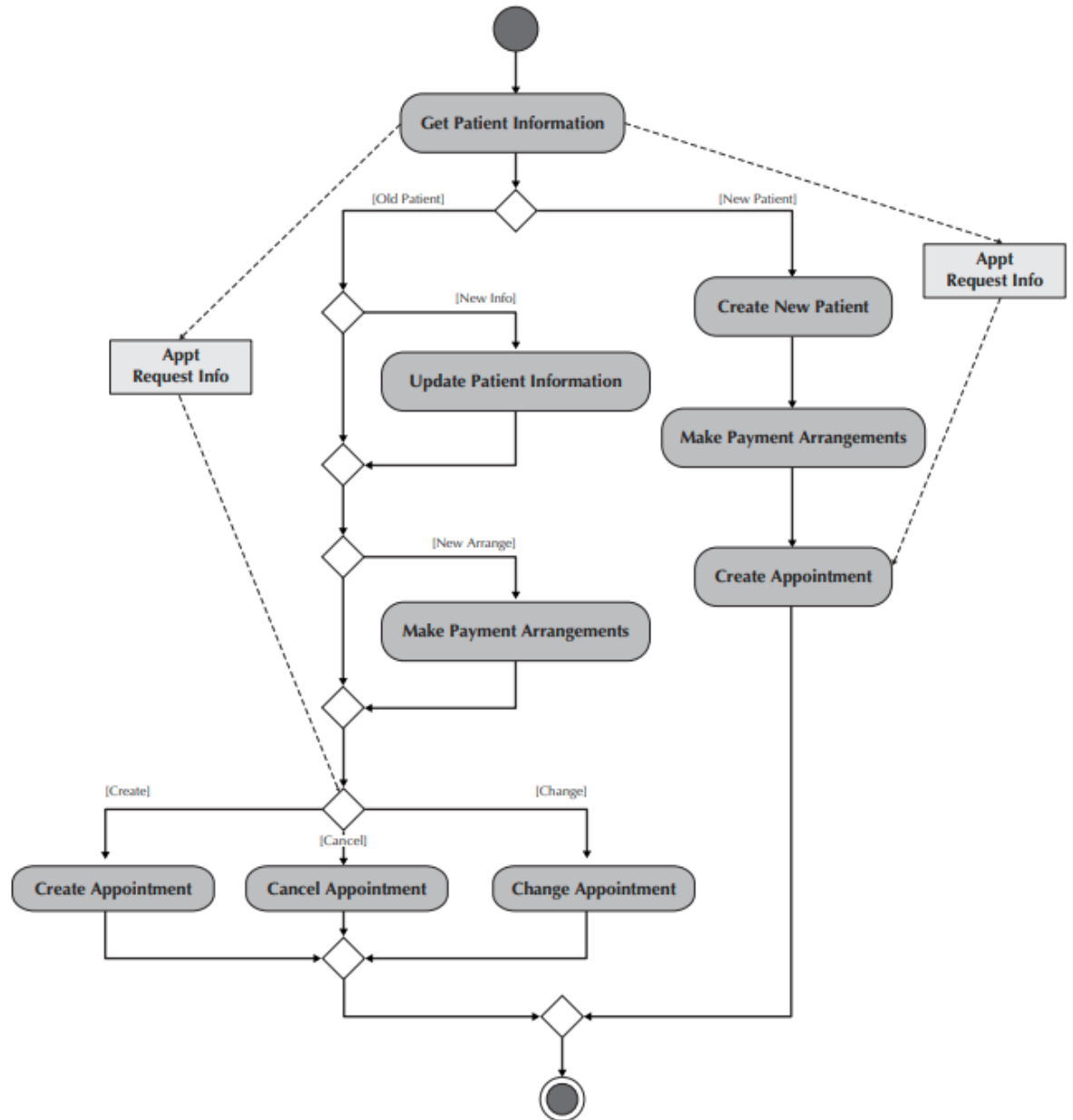


FIGURE 4-8 Activity Diagram for the Manage Appointments Use Case

# Guidelines for creating activity diagram



- ❖ Because an activity diagram can be used to model any kind of process, you should set the context or scope of the activity being modeled.
- ❖ You must identify the activities, control flows, and object flows that occur between the activities.

# Guidelines for creating activity diagram



- ❖ You should identify any decisions that are part of the process being modeled.
- ❖ You should attempt to identify any prospects for parallelism in the process.
- ❖ You should draw the activity diagram.

# Summary



- ❖ Explain how to model a specific use case with an activity diagram.
- ❖ Describe the different elements of an activity diagram.
- ❖ Create an activity diagram that represents a specific use case.



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Q&A